J.A.R.V.I.S. Android Voice Assistant

Complete Technical Documentation, Command Reference & Version 2 Roadmap

Target Device: iQOO 12 (Funtouch OS 14 / Android 14)

Architecture: 100% On-Device / Offline Capable

Status: Fully Built, Tested & Deployed

Voice Engine: Natural British Male (Jarvis Profile)

1. Executive Summary & Architecture

JARVIS is a high-performance, lightweight, privacy-focused voice assistant built specifically for Android. Unlike cloud-dependent assistants, JARVIS operates 100% on-device without cloud latency, subscription fees, or external API costs. It provides instant hands-free control over phone hardware, messaging, navigation gestures, phone calls, media playback, and web queries.

- **JarvisAssistActivity:** Registered in Manifest with android.intent.action.ASSIST. Holding the device Power button launches this activity instantly and silently without verbal interruption.
- **JarvisOverlayService:** Manages the sleek, compact Siri-style bottom floating pill HUD (~120dp height). Features real-time fluid neon sine wave audio visualization reacting to microphone volume levels (rmsLevel).
- **VoiceListeningService:** Foreground background service continuously listening for wake-word triggers ('Hey Jarvis', 'Jarvis') with auto-rebind lifecycle resilience.
- **CommandParser:** Pure rule-based parsing engine. Features compiled regex pattern matching, N-best speech hypothesis scanning (parseCandidates), and intelligent homophone normalizations.
- **JarvisTts:** Custom Text-to-Speech manager configured for a cinematic British male butler voice (en-gb-x-rjs profile at pitch 0.90 / rate 0.98) with automatic speech text sanitization.
- **AppLauncher:** Multi-tiered fuzzy app matching with 60-second metadata caching to resolve exact, partial, and nickname app aliases in under 1 millisecond.
- **TelephonyController:** Manages contact fuzzy search (handling emojis and partial names), direct cellular calling with auto-speakerphone routing, WhatsApp VoIP calling, and silent incoming call ringer control.
- **ActionExecutor:** Central routing module connecting parsed commands with underlying Android OS APIs, Accessibility Service gestures, and system controllers.

2. Required Permissions & System Setup Guide

To enable JARVIS to operate to its fullest capability across all apps, home screen, and lock screen, the following system permissions must be granted:

Permission / Setting	Type	Purpose / Function
Default Digital Assistant	System Setting	Allows holding Power button to trigger JARVIS silently.
Display Over Other Apps	System Overlay	Renders the compact Siri-style floating HUD over any app.
Record Audio	Runtime	Captures microphone input for voice command recognition.
Read Contacts	Runtime	Enables fuzzy contact matching for phone & WhatsApp calls.
Make / Answer Calls	Runtime	Allows placing phone calls and answering/declining incoming calls.
Accessibility Service	Accessibility	Executes touch gestures: Go Back, Home, Recents, Reel Scrolling, Tap.

Battery Optimization	Exclusion	Prevents Android OS from killing background wake-word listening.

3. Comprehensive Command Reference

Below is the complete dictionary of supported voice commands, trigger phrases, homophone fallbacks, and exact actions executed by JARVIS:

Category & Spoken Command	Action Executed & Functionality
--- Activation & Wake Up ---	
Hey Jarvis / Jarvis	Triggers JARVIS background wake-word listener.
Hold Power Button	Triggers JARVIS instantly & silently with bottom Siri floating HUD and haptic tick.
--- App Control ---	
Open [App Name]	Launches any app (e.g. 'Open Chrome', 'Open WhatsApp', 'Open Settings').
Open [Nickname]	Launches via alias (e.g. 'Insta' -> Instagram, 'YT' -> YouTube, 'FB' -> Facebook).
Open YouTube on Brave	Launches browser directly to specific website or search query.
--- Phone Calls & Speakerphone ---	
Call [Contact Name]	Searches contacts with fuzzy match and places cellular call.
Call [Name] on speaker	Initiates call and automatically activates speakerphone audio.
Call [Name] on WhatsApp	Searches WhatsApp contacts and opens direct WhatsApp VoIP call/chat.
--- Incoming Call Control ---	
Accept / Answer	Answers the incoming call immediately.
Decline / Reject	Rejects/ends the incoming call.
Silence / Silent call	Silences phone ringer without rejecting call.
--- Clipboard & Web ---	
Open the link I copied	Reads clipboard text and opens URL or search query directly in Chrome.
Search for [query]	Performs Google Web Search for specified query.
Open [website url]	Navigates directly to specified URL.
--- Navigation & Gesture Control ---	
Go back / Back	Executes back gesture (requires Accessibility).
Go home / Home	Navigates to home screen.
Recent apps / Recents	Opens Android recents task switcher.
Scroll up / Scroll down	Performs smooth scroll up or down on current screen.
Next reel / Previous reel	Swipes to next/previous short video on Reels / Shorts / TikTok.
Like reel / Like	Double taps center of screen to like current reel/post.
--- Volume & Ringer ---	
Volume up / Volume down	Increases or decreases media volume.
Set volume to [0-15]	Sets media volume directly to specific level (0 to 15 or max/min).
Mute / Unmute	Mutes or unmutes media volume.
Silence ringer / Silence	Enables silent ringer mode.
Do not disturb / DND	Toggles Do Not Disturb mode.

--- Media Playback ---	
Play / Pause / Resume	Toggles media playback (Spotify, YouTube, Music).
Next song / Skip	Skips to next track.
Previous song / Last song	Returns to previous track.
--- Flashlight & Utilities ---	
Torch on / Flashlight on	Turns device flashlight ON.
Torch off / Flashlight off	Turns device flashlight OFF.
Toggle flashlight	Toggles flashlight state.
--- Date, Time & Battery ---	
What time is it / Time now	Announces current time in natural voice.
What day is today / Date	Announces today's day and date.
Battery level / Status	Announces current battery percentage and charging state.
--- Alarms & Timers ---	
Set timer for [N] mins/secs	Creates a countdown timer in Clock app.
Wake me up after [N] mins	Sets countdown timer labeled 'Wake up timer'.
Set alarm at [8pm / 7am]	Creates alarm at specified time in Clock app.
Wake me up at [8pm]	Sets alarm at specified time for waking up.

4. Version 2.0 (v2) Strategic Roadmap

To evolve JARVIS into an even smarter, autonomous personal assistant, Version 2.0 will introduce advanced AI, context memory, and deep OS integrations:

- **1. On-Device Local LLM Integration (Small Language Model):** Integrate lightweight offline GGUF model (e.g. Qwen2.5 0.5B / SmolLM 135M via llama.cpp) to handle conversational reasoning, text summarization, and complex multi-step user queries without any internet connection.
- **2. Multi-Turn Context & Preference Memory:** Maintain local SQLite memory database to remember user preferences (e.g., favorite contacts, preferred browser, usual wake-up time, custom shortcuts) across command turns.
- **3. Smart Macro & Custom Routine Engine:** Allow users to define custom triggers and action chains (e.g., 'Leaving Home' -> turns off Wi-Fi, sets media volume to max, launches Google Maps navigation).
- **4. Intelligent Notification Reader & Summarizer:** Read incoming WhatsApp messages and system notifications aloud when requested (e.g., 'Read my unread messages', 'Summarize notifications').
- **5. Screen Vision & Image Context (VLM):** Capture current screen bitmap via Projection API and answer visual queries (e.g. 'What is on my screen?', 'Read text from screen', 'Scan QR code').
- **6. Deep Media & Music Automation:** Direct voice search and playback inside Spotify and YouTube Music (e.g. 'Play A.R. Rahman on Spotify').
- **7. Smart Home (IoT) Integration:** Local MQTT / Home Assistant integration to control smart lights, fans, and switches directly from JARVIS.